

- Seyed Iman Mirzadeh, Keivan Alizadeh, Hooman Shahrokhi, Oncel Tuzel, Samy Bengio, and Mehrdad Farajtabar. GSM-symbolic: Understanding the limitations of mathematical reasoning in large language models. In *The Thirteenth International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=AjXkRZIVjB>.
- OpenAI. Openai o1 system card, 2024. URL <https://openai.com/index/openai-o1-system-card/>.
- OpenAI. Introducing gpt-5.2, 2025. URL <https://openai.com/index/introducing-gpt-5-2/>.
- OpenAI. Gpt-5.4 thinking system card. <https://deploymentsafety.openai.com/gpt-5-4-thinking/gpt-5-4-thinking.pdf>, 2026.
- Yotam Perlitz, Elron Bandel, Ariel Gera, Ofir Arviv, Liat Ein-Dor, Eyal Shnarch, Noam Slonim, Michal Shmueli-Scheuer, and Leshem Choshen. Efficient benchmarking (of language models). In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pp. 2519–2536, 2024.
- David Rein, Betty Li Hou, Asa Cooper Stickland, Jackson Petty, Richard Yuanzhe Pang, Julien Dirani, Julian Michael, and Samuel R. Bowman. GPQA: A graduate-level google-proof q&a benchmark. In *First Conference on Language Modeling*, 2024. URL <https://openreview.net/forum?id=Ti67584b98>.
- Reuven Rubinstein. The cross-entropy method for combinatorial and continuous optimization. *Methodology and computing in applied probability*, 1(2):127–190, 1999.
- Reuven Y Rubinstein and Dirk P Kroese. *The cross-entropy method: a unified approach to combinatorial optimization, Monte-Carlo simulation, and machine learning*, volume 133. Springer, 2004.
- Keisuke Sakaguchi, Ronan Le Bras, Chandra Bhagavatula, and Yejin Choi. Winogrande: An adversarial winograd schema challenge at scale. *arXiv preprint arXiv:1907.10641*, 2019.
- Sang T. Truong, Yuheng Tu, Percy Liang, Bo Li, and Sanmi Koyejo. Reliable and efficient amortized model-based evaluation. In *Forty-second International Conference on Machine Learning*, 2025. URL <https://openreview.net/forum?id=HDbWrsgkB9>.
- Joshua Vendrow, Edward Vendrow, Sara Beery, and Aleksander Madry. Large language model benchmarks do not test reliability. In *Neurips Safe Generative AI Workshop 2024*, 2024. URL <https://openreview.net/forum?id=XSeN6xZtZ9>.
- Rajan Vivek, Kawin Ethayarajh, Diyi Yang, and Douwe Kiela. Anchor points: Benchmarking models with much fewer examples. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 1576–1601, 2024.
- Alex Wang, Amanpreet Singh, Julian Michael, Felix Hill, Omer Levy, and Samuel Bowman. GLUE: A multi-task benchmark and analysis platform for natural language understanding. In Tal Linzen, Grzegorz Chrupała, and Afra Alishahi (eds.), *Proceedings of the 2018 EMNLP Workshop BlackboxNLP: Analyzing and Interpreting Neural Networks for NLP*, pp. 353–355, Brussels, Belgium, November 2018. Association for Computational Linguistics. doi: 10.18653/v1/W18-5446. URL <https://aclanthology.org/W18-5446/>.
- Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc V Le, Ed H. Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. In *The Eleventh International Conference on Learning Representations*, 2023. URL <https://openreview.net/forum?id=1PL1NIMMrw>.
- Skyler Wu, Yash Nair, and Emmanuel J. Candès. Efficient evaluation of llm performance with statistical guarantees, 2026. URL <https://arxiv.org/abs/2601.20251>.

An Yang, Beichen Zhang, Binyuan Hui, Bofei Gao, Bowen Yu, Chengpeng Li, Dayiheng Liu, Jianhong Tu, Jingren Zhou, Junyang Lin, Keming Lu, Mingfeng Xue, Runji Lin, Tianyu Liu, Xingzhang Ren, and Zhenru Zhang. Qwen2.5-math technical report: Toward mathematical expert model via self-improvement. *arXiv preprint arXiv:2409.12122*, 2024.

Xiang Yue, Yuansheng Ni, Kai Zhang, Tianyu Zheng, Ruoqi Liu, Ge Zhang, Samuel Stevens, Dongfu Jiang, Weiming Ren, Yuxuan Sun, Cong Wei, Botao Yu, Ruibin Yuan, Renliang Sun, Ming Yin, Boyuan Zheng, Zhenzhu Yang, Yibo Liu, Wenhao Huang, Huan Sun, Yu Su, and Wenhui Chen. Mmmu: A massive multi-discipline multimodal understanding and reasoning benchmark for expert agi. In *Proceedings of CVPR*, 2024.

Xiang Yue, Tianyu Zheng, Yuansheng Ni, Yubo Wang, Kai Zhang, Shengbang Tong, Yuxuan Sun, Botao Yu, Ge Zhang, Huan Sun, Yu Su, Wenhui Chen, and Graham Neubig. Mmmu-pro: A more robust multi-discipline multimodal understanding benchmark. In *Proceedings of ACL*, 2025.

Rowan Zellers, Ari Holtzman, Yonatan Bisk, Ali Farhadi, and Yejin Choi. Hellaswag: Can a machine really finish your sentence? In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, 2019.

A LLM Inference Setup

We describe the implementation details for LLM inference. We use the same system prompt for all models:

Please reason step by step, and put your final answer within `\\boxed{}`.

and parse the final answer within the box. We follow the recommended decoding hyperparameters specified in each model card: temperature of 0.7 and top-p of 0.8 for Qwen2.5-Math-7B-Instruct, 1.0 and 1.0 for gpt-oss-20b-low, and 1.0 and 0.95 for Gemini 2.5 Flash Lite. For all models, we use a maximum of 32768 new tokens. If not specified, we use the default values.

B Algorithm for calculating confidence bounds

Algorithm 2 COMPUTEESTIMATE: Importance Sampling with Confidence Bounds

- 1: **Input:** Failure indicator f , distributions P and Q , sample size N , confidence level α
 - 2: **Output:** Estimate $\hat{\mu}$, confidence interval $[\hat{\mu}_{\text{low}}, \hat{\mu}_{\text{high}}]$
 - 3: Sample $Z_1, \dots, Z_N \sim Q$
 - 4: Compute importance-weighted estimate: $\hat{\mu} \leftarrow \frac{1}{N} \sum_{i=1}^N f(Z_i) \frac{P(Z_i)}{Q(Z_i)}$
 - 5: Compute sample variance: $\hat{\sigma}^2 \leftarrow \frac{1}{N-1} \sum_{i=1}^N \left(f(Z_i) \frac{P(Z_i)}{Q(Z_i)} - \hat{\mu} \right)^2$
 - 6: Set $Z_{\alpha/2} \leftarrow$ critical value from standard normal for confidence level α
 - 7: Compute CI: $[\hat{\mu}_{\text{low}}, \hat{\mu}_{\text{high}}] \leftarrow \left[\hat{\mu} - Z_{\alpha/2} \frac{\hat{\sigma}}{\sqrt{N}}, \hat{\mu} + Z_{\alpha/2} \frac{\hat{\sigma}}{\sqrt{N}} \right]$
 - 8: **Return** $\hat{\mu}, [\hat{\mu}_{\text{low}}, \hat{\mu}_{\text{high}}]$
-

Algorithm 2 details the COMPUTEESTIMATE subroutine called in Algorithm 1 (Section 5).

C Model-Template Pairs with Too Few Failures

The following model-template pairs have zero observed failures within our 100K evaluation budget to support reliable estimation (failure counts from the full 100K cache):

- gpt-oss-20b-low: Templates 1, 3, 5, 6 ($K = 8$)
- Gemini 2.5 Flash Lite: Templates 1, 3, 5, 7, 8 ($K = 8$)

We expect systematic failure patterns may emerge with a larger evaluation budget, and leave this for future work.